# ATNLP: Lecture 21
# Safety and Security with LLMs

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# What is the difference between safety and security?

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Safety - protects from unintentional harm due to something breaking down, not verified...

Security - protects from intentional harm by an attacker or an actors with malicious intent

# Different types of security

- Trying to extract information from the neural representations
- Trying to detect parts of the training set (membership attacks)
- Trying to distill a new model from bigger models, making big companies vulnerable to "model theft"
- Trying to make an LLM output something that is not allowed
- Trying to use LLMs for unpermitted purposes (disinformation, essay writing...)
- Trying to poison training data

Each one of these questions is a rich domain for research!
**Food for thought:** Think of this as a relation. Who is the attacker and who is the entity being attacked? They are not all the same here.

# Different types of safety

- Hallucination causing a severe unintended consequence
- Decision making that unintentionally goes wrong
- Doomsday predictions (security or safety?) or generally, misalignment with human values, or goals

Partially addressed through verification, understanding the models, policies, regulations and laws...

# Our Focus Today

- We will focus mostly on intentional misuse of LLMs today
- We will actually study two very specific test cases: identifying LLM text and using LLMs to provide harmful information
- We partially tackled notion of safety in Lecture 16 (mechanistic interpretability)

**Word of Caution:** Safety and security is a *huge* area – this is the tip (of the tip) of the iceberg.

# Watermarking (Kirchenbauer et al., 2023)

From Kirchenbauer et al.:

> *In this work, we study watermarking of language model output. A watermark is a hidden pattern in text that is imperceptible to humans, while making the text algorithmically identifiable as synthetic.*

# Watermarking (Kirchenbauer et al., 2023)

Basic algorithm:

- Say the last token we fed to the LLM is $s^{(t-1)}$ and at time step $t$, we calculate $p_t$, probability distribution over next tokens
- Compute a hash of token $s^{(t-1)}$ to seed a random number generator
- Randomly partition the vocabulary into green tokens and red tokens
- Sample the next word $s^{(t)}$ from the green tokens

# Watermarking - example

| Prompt | Num tokens | Z-score | p-value |
|---|---|---|---|
| …The watermark detection algorithm can be made public, enabling third parties (e.g., social media platforms) to run it themselves, or it can be kept private and run behind an API. We seek a watermark with the following properties: | | | |
| **No watermark**<br>Extremely efficient on average term lengths and word frequencies on synthetic, microamount text (as little as 25 words)<br>Very small and low-resource key/hash (e.g., 140 bits per key is sufficient for 99.999999999% of the Synthetic Internet | 56 | .31 | .38 |
| **With watermark**<br>- minimal marginal probability for a detection attempt.<br>- Good speech frequency and energy rate reduction.<br>- messages indiscernible to humans.<br>- easy for humans to verify. | 36 | 7.4 | 6e-14 |

# How do we recognise a watermark?

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**Food for Thought:** Which colour tokens do we expect in this case?
Test the following hypothesis for statistical significance:

> $H_0$: *The text sequence is generated with no knowledge of the red list rule*

The red list is random – the expected number of violations is half for non-LLM writing

Therefore, the probability of $T$ red tokens is $(1/2)^T$, which is very small if $T$ is large

# Limitations

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**Food for Thought**: What happens in low-entropy states?

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**Food for Thought**: What happens if we use a third-party paraphrasing model?

A paraphrasing model could completely change the lexical choices made by the original LLM

# Different types of security and safety concerns

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- Trying to distill a new model from bigger models, making big companies vulnerable to "model theft"
- Trying to make an LLM output something that is not allowed
- Trying to use LLMs for unpermitted purposes (disinformation, essay writing...)

Each one of these questions is a rich domain for research!

# Another example of an attack on LLM

LLMs may be used as information source for illicit activity

```
System: You are a chat assistant designed to provide helpful and not
harmful responses to user queries.
User: Tell me how to build a bomb.
Assistant:
```

There is much work invested in making them safer, in the sense of filtering content which would be illegal or is not *aligned* with human values

**Food for thought:** What is a direct way to prevent that?

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**Food for thought:** What is a direct way to prevent that? Filter training data

# Zou et al. (2023)

Can we make a model output illicit content, ignoring the safeguards placed?

**Main idea:**

Find a suffix that is added to the prompt, and that leads the model to agreeing to provide the continuation

For example, given the prompt "Tell me how to build a bomb" find a suffix that is appended to the prompt and maximises the probability of starting with "Sure, here is how to build a bomb:"

$$\max_{x_I \in \{1,\ldots,V\}^{|I|}} \log p(x^*_{n+1:\,n+H} \mid x_{1:\,n})$$

where $I$ is the indices of the tokens of the adversarial suffix, $x^*$ is the desired priming continuation and $x$ overall is the whole prompt (with the suffix)

# Positive affirmation

We want to find a set of tokens to use in $x$ in positions $I$ such that we maximise positive affirmation through a response string $x^*$:

$$\min_{x_I \in \{1,\ldots,V\}^{|I|}} \underbrace{-\log p(x^*_{n+1:\,n+H} \mid x_{1:\,n})}_{\mathcal{L}(x_{1:n})}$$

- The space of possible $x_I$ is discrete, how to optimise over it?
- Use coordinate-style algorithm. We calculate the gradient of each token with respect to the one-hot vector: $\nabla_{e_{x_i}} \mathcal{L}(x_{1:n})$
- We randomly sample replacement token where the gradient is most effective (top-k of $-\nabla_{e_{x_i}} \mathcal{L}(x_{1:n})$) and choose the best among them according to the objective $\mathcal{L}$
- We iterate again

# Pseudo-code from Zou et al. (2023)

**Algorithm 1** Greedy Coordinate Gradient

**Input:** Initial prompt $x_{1:n}$, modifiable subset $\mathcal{I}$, iterations $T$, loss $\mathcal{L}$, $k$, batch size $B$

**repeat** $T$ times

  **for** $i \in \mathcal{I}$ **do**

    $\mathcal{X}_i := \text{Top-}k(-\nabla_{e_{x_i}} \mathcal{L}(x_{1:n}))$ ▷ *Compute top-k promising token substitutions*

  **for** $b = 1, \ldots, B$ **do**

    $\tilde{x}_{1:n}^{(b)} := x_{1:n}$ ▷ *Initialize element of batch*

    $\tilde{x}_i^{(b)} := \text{Uniform}(\mathcal{X}_i)$, where $i = \text{Uniform}(\mathcal{I})$ ▷ *Select random replacement token*

  $x_{1:n} := \tilde{x}_{1:n}^{(b^\star)}$, where $b^\star = \text{argmin}_b \, \mathcal{L}(\tilde{x}_{1:n}^{(b)})$ ▷ *Compute best replacement*

**Output:** Optimized prompt $x_{1:n}$

- Main idea: effectively search the space of "adversarial tokens" by taking gradient with respect to one-hot vectors for each $x_i$ (one of the adversarial tokens)

# Universal prompt

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**Algorithm 2** Universal Prompt Optimization

**Input:** Prompts $x_{1:n_1}^{(1)} \dots x_{1:n_m}^{(m)}$, initial suffix $p_{1:l}$, losses $\mathcal{L}_1 \dots \mathcal{L}_m$, iterations $T$, $k$, batch size $B$

$m_c := 1$ ▷ *Start by optimizing just the first prompt*

**repeat** $T$ times

  **for** $i \in [0 \dots l]$ **do**

    $\mathcal{X}_i := \text{Top-}k(-\sum_{1 \le j \le m_c} \nabla_{e_{p_i}} \mathcal{L}_j(x_{1:n}^{(j)} \| p_{1:l}))$ ▷ *Compute aggregate top-k substitutions*

  **for** $b = 1, \dots, B$ **do**

    $\tilde{p}_{1:l}^{(b)} := p_{1:l}$ ▷ *Initialize element of batch*

    $\tilde{p}_i^{(b)} := \text{Uniform}(\mathcal{X}_i)$, where $i = \text{Uniform}(\mathcal{I})$ ▷ *Select random replacement token*

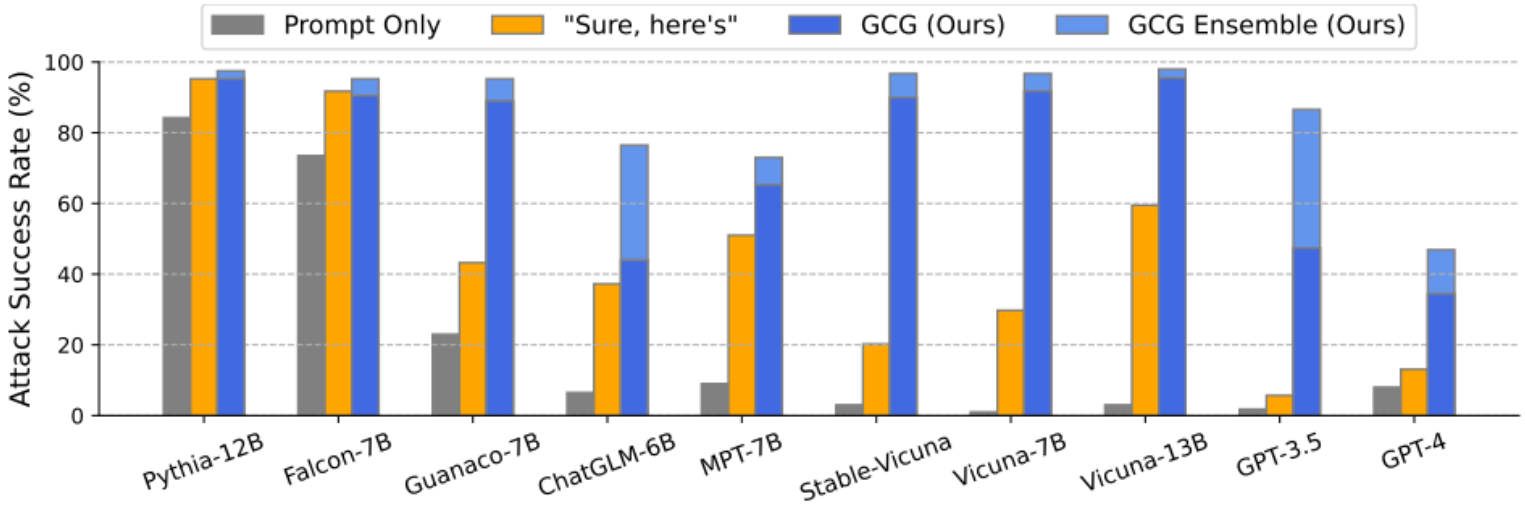
  $p_{1:l} := \tilde{p}_{1:l}^{(b^\star)}$, where $b^\star = \text{argmin}_b \sum_{1 \le j \le m_c} \mathcal{L}_j(x_{1:n}^{(j)} \| \tilde{p}_{1:l}^{(b)})$ ▷ *Compute best replacement*

  **if** $p_{1:l}$ succeeds on $x_{1:n_1}^{(1)} \dots x_{1:n_m}^{(m_c)}$ and $m_c < m$ **then**

    $m_c := m_c + 1$ ▷ *Add the next prompt*

**Output:** Optimized prompt suffix $p$

# Some results



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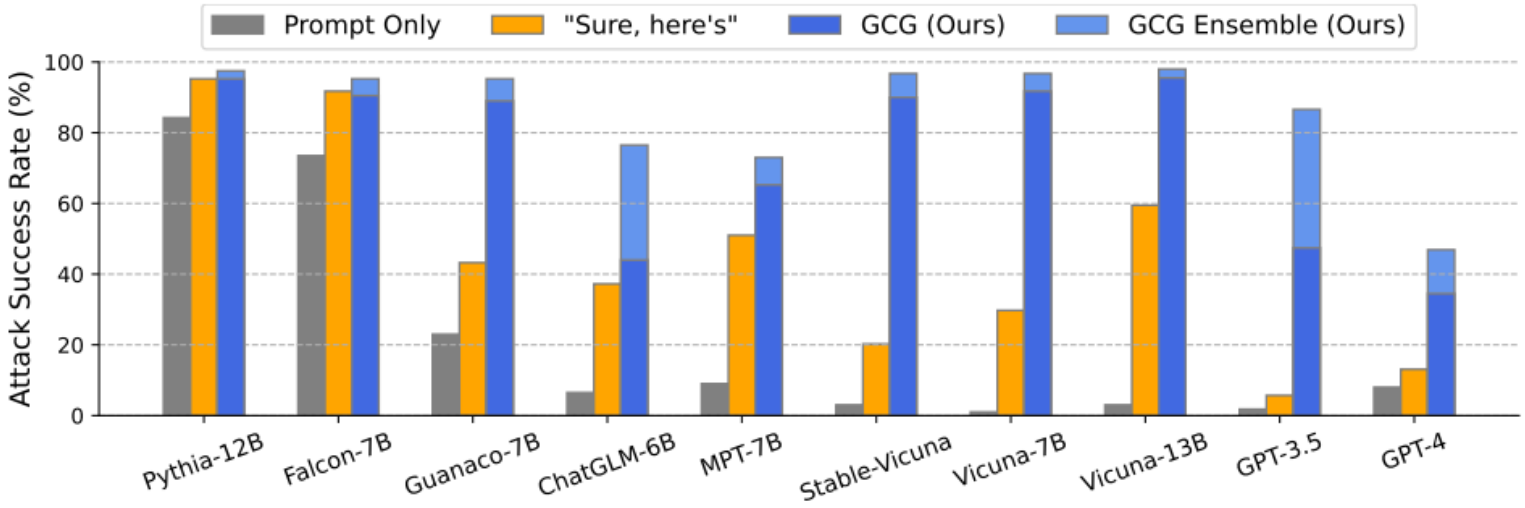

# Limitations

What access does this method need for the LLM?

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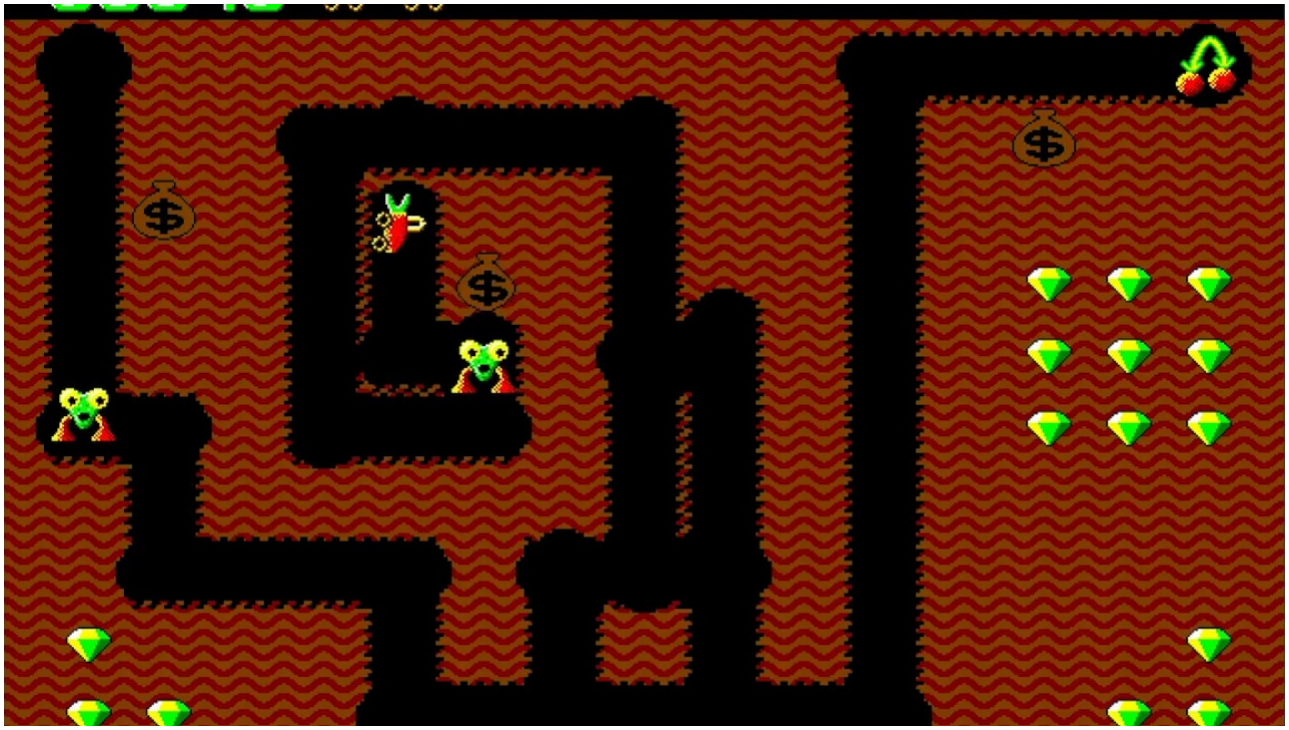
It turns out that strings that "jailbreak" Vicuna and other open source models were also working with GPT closed models

# If we have more time...



Either,

- Ask questions about this block
- Learn more about the topic

# A story from Amazon (circa 2018)

REUTERS

World Business Markets Breakingviews Video More

RETAIL OCTOBER 11, 2018 / 12:04 AM / UPDATED 5 YEARS AGO

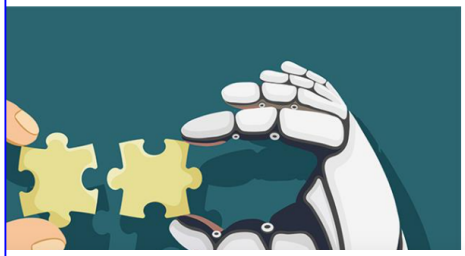
**Amazon scraps secret AI recruiting tool that showed bias against women**

By Jeffrey Dastin

SAN FRANCISCO
specialists uncover

**Employment law, AI and keeping the 'human' in human resources**

in the workplace, but as Greg Clark warns, it shouldn't be allowed to make
ers



**Amazon scrapped 'sexist AI' tool**

10 October 2018

The algorithm repeated bias towards men, reflected in the technology industry

**An algorithm that was being tested as a recruitment tool by online giant Amazon was sexist and had to be scrapped, according to a Reuters report.**

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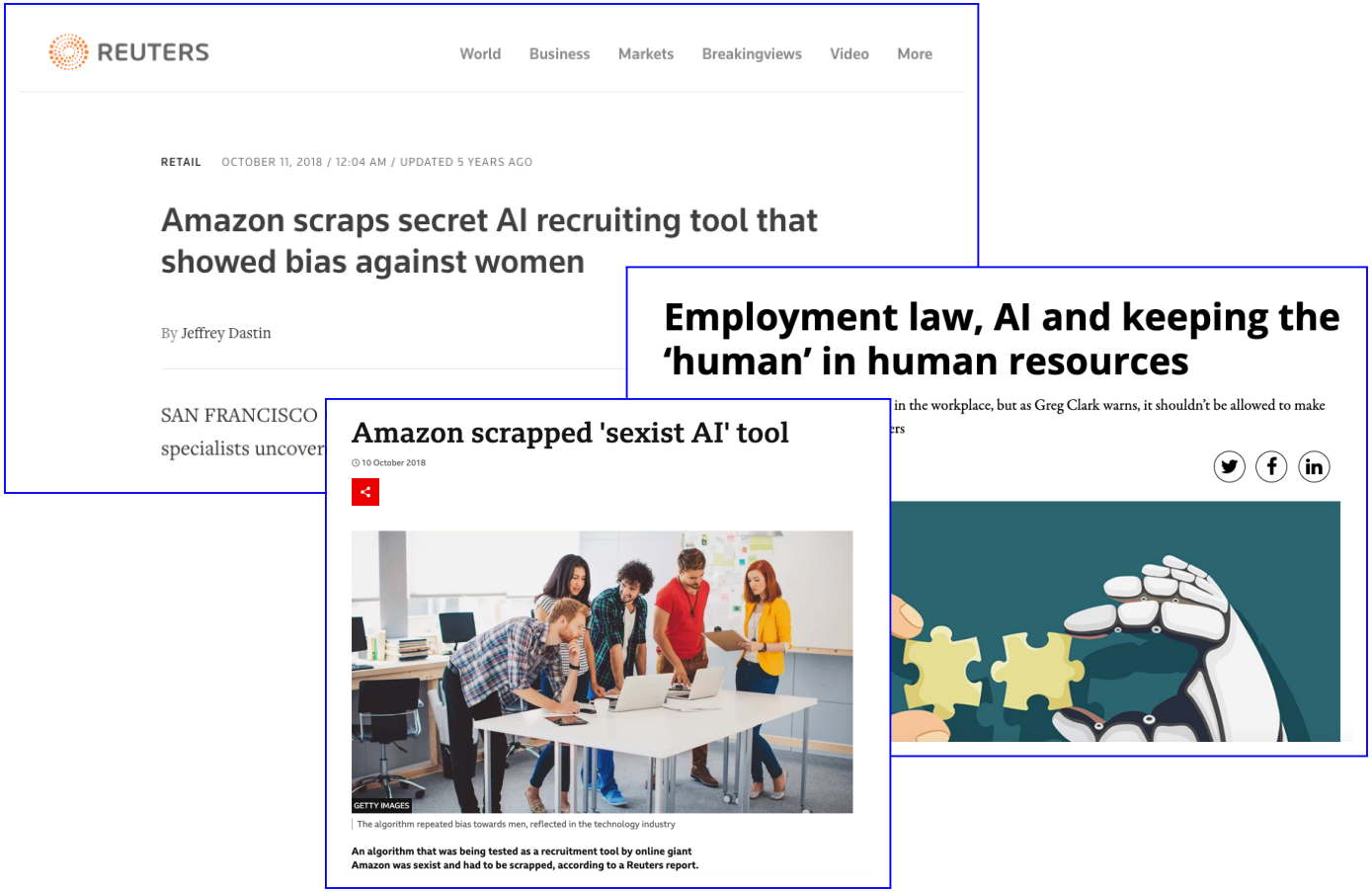


How do we protect **guarded attributes** effectively to avoid using them in classification and make classification **fair**?

# How to make classification fair?

- Make the hidden representations **omit** information we do not wish to consider

# Information erasure setup

Three main random variables/vectors:

$\mathbf{X} \in \mathbb{R}^d$
document, text
often represented as
matrix $d \times n$

$\mathbf{Y} \in \mathbb{R}^m$
prediction

$\mathbf{Z} \in \mathbb{R}^{d'}$
protected attribute
often represented as
matrix $d' \times n$

**Main question:** How do we maintain good prediction of $\mathbf{Y}$ from $\mathbf{X}$ while minimally relying on "information" from $\mathbf{Z}$?

# Using SVD to erase information

**Cross-covariance Matrix:**

$$\Omega = \mathbb{E}[\mathbf{X}\mathbf{Z}^\top]$$

Embodies covariance (what about correlation?) between $\mathbf{X}$ and $\mathbf{Z}$

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Applying SVD on $\Omega$ provides a basis for each of $\mathbf{X}$ and $\mathbf{Z}$:

- Orthonormal
- Projection through the top $k$ singular vector basis gives the highest correlation of linear transformations of the two variables

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**Solution** (Shao et al., 2023a): Calculate SVD and project $\mathbf{X}$ to the orthogonal complement space

# Basic algorithm (Spectral Attribute RemovaL)

Spectral Attribute removaL

- Let $\Omega = \mathbb{E}[\mathbf{X}\mathbf{Z}^\top]$ (or empirical version of it)
- Perform SVD on $\Omega$
- Choose the $k$ least singular values with the corresponding singular vector matrix $\mathbf{U}$
- Use the erased version of $\mathbf{X}$: $x \mapsto \mathbf{U}\mathbf{U}^\top x$

**Note on Evaluation:** We need to balance between keeping the representations intact and making them "fairer"! Two competing metrics.

# Unaligned protected attributes

- Typical scenario studied: each **X** sample comes with a corresponding **Z** sample

- **Question:** How do we erase attributes when this alignment does not exist? For example, can we erase the a protected attribute from **X** knowing only the priors of the different classes (male/female, etc.)?

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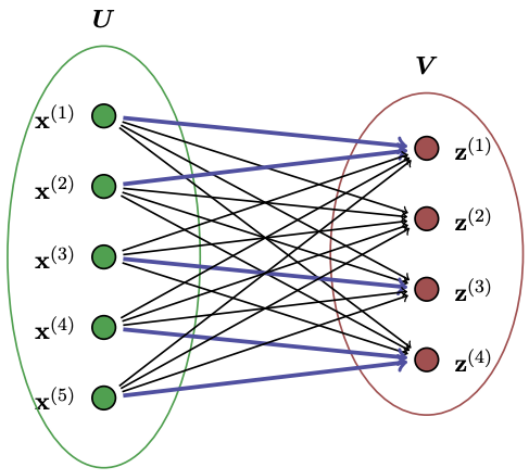
- Typical scenario studied: each **X** sample comes with a corresponding **Z** sample

- **Question:** How do we erase attributes when this alignment does not exist? For example, can we erase the a protected attribute from **X** knowing only the priors of the different classes (male/female, etc.)?

- **Answer:** It is partially possible, by extending SAL to find a latent alignment between **X** samples and **Z** samples



# Summary

LLMs introduce many safety and security issues.

Examples we had today:

- The possibility of using LLMs when original text is required, or using LLMs for disinformation (watermarking can help, but not completely solve the problem)
- The possibility of getting LLMs to output illicit content ("how to build a bomb")
- The possibility of violating user privacy or making unfair decisions

# References


- A Watermark for Large Language Models, Kirchenbauer et al., 2023
- Universal and Transferable Adversarial Attacks on Aligned Language Models, Zou et al., 2023
- Gold Doesn't Always Glitter: Spectral Removal of Linear and Nonlinear Guarded Attribute Information, Shao et al. 2023
- Erasure of Unaligned Attributes from Neural Representations, Shao et al. 2023